

Deep Learning

Assignment 2

Implement Simple and Multiple Linear Regression to predict continuous variables.

1. Perform data preprocessing (handle missing values, feature scaling).
2. Fit a Simple Linear Regression model on a dataset (e.g., predicting house prices).
3. Extend to Multiple Linear Regression with multiple features.
4. Evaluate models using MSE, RMSE, and R^2 Score.

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_squared_error, r2_score
```

```
df = pd.read_csv('insurance.csv')
```

```
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1338 non-null   int64
1   sex         1338 non-null   object
2   bmi         1338 non-null   float64
3   children    1338 non-null   int64
4   smoker      1338 non-null   object
5   region      1338 non-null   object
6   charges     1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```
df.describe()
```

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

```
print("\nMissing values:\n", df.isnull().sum())
```

Missing values:

```
age      0
sex      0
bmi      0
children 0
smoker   0
region   0
charges  0
dtype: int64
```

```
X = df.drop('charges', axis=1)
y = df['charges']
```

```
cat_cols = ['sex', 'smoker', 'region']
num_cols = ['age', 'bmi', 'children']
```

```
numeric_transformer = Pipeline(steps=[
    ('scaler', StandardScaler())
])

categorical_transformer = Pipeline(steps=[
    ('onehot', OneHotEncoder(drop='first'))
])

preprocessor = ColumnTransformer(
    transformers=[
        ('num', numeric_transformer, num_cols),
        ('cat', categorical_transformer, cat_cols)
    ])

```

```
X_simple = df[['bmi']] # single feature
X_train_s, X_test_s, y_train_s, y_test_s = train_test_split(X_simple, y, test_size=0.2, random_state=42)

simple_model = LinearRegression()
simple_model.fit(X_train_s, y_train_s)

y_pred_s = simple_model.predict(X_test_s)
```

```
multi_model = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('regressor', LinearRegression())
])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

multi_model.fit(X_train, y_train)
y_pred_m = multi_model.predict(X_test)
```

```
# Simple Linear Regression
mse_simple = mean_squared_error(y_test_s, y_pred_s)
rmse_simple = np.sqrt(mse_simple)
r2_simple = r2_score(y_test_s, y_pred_s)
```

```
print("Simple Linear Regression")
print("MSE:", mse_simple)
print("RMSE:", rmse_simple)
print("R2 Score:", r2_simple)
```

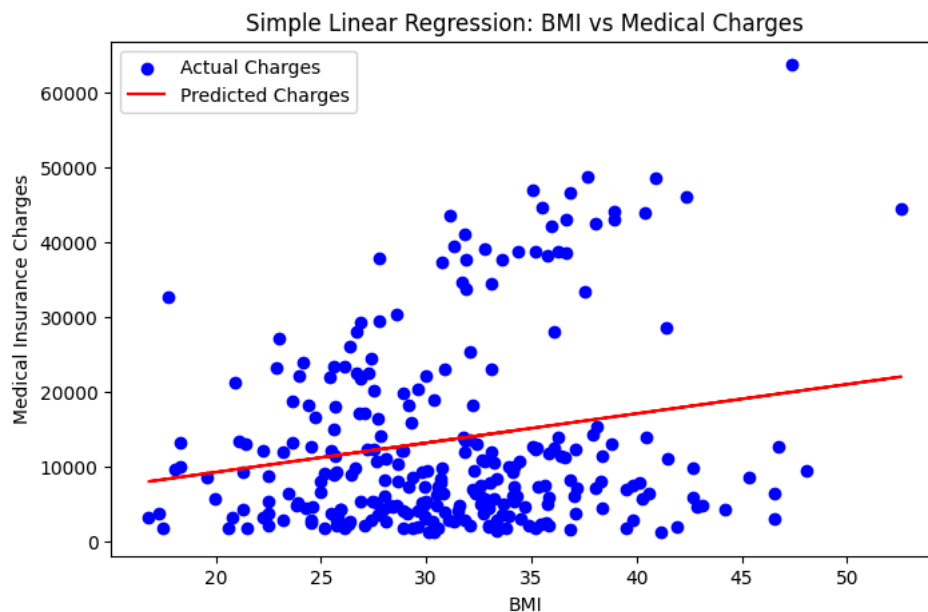
```
Simple Linear Regression
MSE: 149085057.03839505
RMSE: 12210.039190698571
R2 Score: 0.03970193117941878
```

```
# Multiple Linear Regression
mse_multi = mean_squared_error(y_test, y_pred_m)
rmse_multi = np.sqrt(mse_multi)
r2_multi = r2_score(y_test, y_pred_m)

print("\nMultiple Linear Regression")
print("MSE:", mse_multi)
print("RMSE:", rmse_multi)
print("R2 Score:", r2_multi)
```

```
Multiple Linear Regression
MSE: 33596915.85136146
RMSE: 5796.2846592762735
R2 Score: 0.7835929767120723
```

```
# Simple Linear Regression Plot
plt.figure(figsize=(8,5))
plt.scatter(X_test_s, y_test_s, color='blue', label='Actual Charges')
plt.plot(X_test_s, y_pred_s, color='red', label='Predicted Charges')
plt.title('Simple Linear Regression: BMI vs Medical Charges')
plt.xlabel('BMI')
plt.ylabel('Medical Insurance Charges')
plt.legend()
plt.show()
```



```
# Multiple Linear Regression: Actual vs Predicted
plt.figure(figsize=(8,5))
plt.scatter(y_test, y_pred_m, color='green')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], color='red')
plt.title('Multiple Linear Regression: Actual vs Predicted Charges')
plt.xlabel('Actual Charges')
plt.ylabel('Predicted Charges')
plt.show()
```

